

Jaeseo Lee

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Summary

I specialize in ensuring software reliability and correctness through formal methods and program analysis. I am particularly interested in applying formal reasoning and analysis to strengthen software quality in large-scale, safety-critical systems.

Work Experience

Samsung Research Staff Engineer in Software Engineering Team	Seoul, South Korea Apr 2026 - Present
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Education

Software Verification Lab. (POSTECH) MS/Ph.D. in Computer Science and Engineering	Pohang, South Korea Feb 2017 - Feb 2026
Pohang University of Science and Technology (POSTECH) BS in Industrial and Management Engineering	Pohang, South Korea Mar 2011 - Feb 2017
University of California, Berkeley Concurrent Enrollment Program	Berkeley, California Jan 2015 – Dec 2015

- Coursework: Operating Systems, Architecture, Machine Learning, Compiler, Security

Industry Collaboration Projects

Verification on PLC Programs , with KSOE (HD Korea Shipbuilding & Offshore Engineering Co., Ltd.)	Jan 2020 – Dec 2020
• Clarified the ambiguous semantics of PLC language described in natural languages • Devised a bounded linear temporal logic (LTL) model checking method that checks conformity of PLC programs to specifications • Designed a specification language for expressing desired properties of PLC programs • Developed STBMC [tool] that integrates the whole process of PLC program verification. This tool generates a counterexample if and only if one exists	
Equivalence of LLVM IR Programs , with GT One	June 2017 – Nov 2018
• Machine-proved semantic equivalence of original and transformed code in security-enhancing transformations • Developed a lightweight tool with a translation validation approach	

Publications

Bounded Model Checking of Multitask PLC ST Programs with Preemption using Rewriting Modulo SMT Jaeseo Lee, Kyungmin Bae [paper]	JLAMP, 2026
Formal Analysis of Networked PLC Controllers Interacting with Physical Environments Jaeseo Lee, Kyungmin Bae [paper]	SAS, 2025
Formal Semantics and Analysis of Multitask PLC ST Programs with Preemption Jaeseo Lee, Kyungmin Bae [paper]	FM, 2024
Bounded Model Checking of PLC ST Programs using Rewriting Modulo SMT Jaeseo Lee, Sangki Kim, Kyungmin Bae [paper]	FTSCS, 2022

Lightweight Equivalence Checking of Code Transformation for Code Pointer Integrity (in Korean)

KCSE, 2019.12

Jaeseo Lee, Tae-Hyoung Choi, Gyuho Lee, Jaegwan Yu, Kyungmin Bae [\[paper\]](#)

Teaching

CSED332: Software Design Methods (TA)

Fall 2017, Fall 2019

CSED321: Programming Languages (TA)

Spring 2019

Scholarships

National Science & Technology Scholarship, by KOSAF (Kr. Student Aid Foundation)

Mar 2011 - Feb 2017